Sparsity-as-Signal: Is it Easier to Test Generalization in Historically-Cutoff Models?

Ari Holtzman and NeuriCo

Abstract

Evaluating generalization in Large Language Models (LLMs) is increasingly difficult due to widespread data contamination in modern web-scale training corpora. We investigate whether models trained on sparse, historically-cutoff data provide a more scientifically rigorous testbed for measuring generalization. Specifically, we test the Sparsity-as-Signal hypothesis: that the lack of modern concepts in a model's training data creates a clear, binary-like signal for generalization when those concepts are introduced via in-context learning. We compare TALKIE-1930, a 13B model with a strict 1930 cutoff, against a modern baseline (TALKIE-WEB) on tasks that were non-existent in 1930, such as Python coding (HUMANEVAL). Our results show that while TALKIE-1930 fails entirely in a ZERO-SHOT setting (0% success), it achieves a 33% success rate on simple logic tasks after just 3 shots, representing a "pure" logic leap. Furthermore, we identify a sharp Temporal Knowledge Cliff in performance on subjects dominated by post-1930 knowledge. We conclude that historically-limited training data simplifies the detection of generalization by eliminating the noise of memorized knowledge, making TALKIE-1930 a valuable instrument for studying the mechanics of machine reasoning.

1 Introduction

Data contamination is the "silent killer" of Large Language Model (LLM) evaluation. As models are trained on increasingly large portions of the internet, finding truly "unseen" tasks to test generalization becomes nearly impossible. When a modern model performs well on a benchmark, it is often difficult to determine if it is genuinely *reasoning* or simply *reciting* examples seen during pretraining. This ambiguity undermines our ability to scientifically measure the mechanics of structural generalization.

Why does this matter? If we cannot distinguish between memorization and reasoning, we cannot accurately assess the progress of AI capabilities or identify the scaling laws of actual intelligence. There is a growing need for a "clean room" environment where modern concepts are guaranteed to be novel to the model.

In this work, we focus on TALKIE-1930 Radford et al. [2026], a 13B parameter model trained exclusively on text published before 1931. This strict temporal cutoff provides a unique, contamination-free environment for testing generalization. We hypothesize the Sparsity-as-Signal principle: that the sparsity and historical limitation of the training data actually make generalization *easier to detect*. Because TALKIE-1930 lacks any exposure to modern technical concepts (e.g., Python programming, DNA, quantum computing), any success it achieves on these tasks must be due to the abstract mapping of its "vintage" knowledge to novel syntaxes.

We empirically test this hypothesis by comparing TALKIE-1930 with a modern dense-data baseline (TALKIE-WEB). We focus on HUMANEVAL Radford et al. [2026] as a primary testbed, given that Python was created in 1991, sixty years after the model's cutoff. Our quantitative preview shows that TALKIE-1930 exhibits a binary-like transition: 0% success in ZERO-SHOT settings, rising to

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33% success on simple logic tasks after just 3 shots. This delta is a high-signal indicator of structural generalization that is often obscured in modern models by high baseline performance.

Our main contributions are as follows:

- We propose and formalize the Sparsity-as-Signal hypothesis, suggesting that data sparsity can be a scientific asset for isolating generalization.
- We empirically demonstrate the Temporal Knowledge Cliff in TALKIE-1930, showing a sharp performance drop-off on post-1930 knowledge which validates the model as a contamination-free instrument.
- We provide a comparative analysis of few-shot gains in coding tasks, showing that TALKIE-1930's "generalization slope" is more discrete and measurable than that of modern models.
- We establish TALKIE-1930 as a robust baseline for cross-era analogical transfer research.

2 Related Work

Our work sits at the intersection of historical language modeling, representation analysis, and the evaluation of LLM reasoning.

Historical Language Models. The release of TALKIE-1930 Radford et al. [2026] represents a shift toward specialized pre-training on curated historical corpora. While most LLMs are evaluated for their modern knowledge, Underwood et al. [2025] argue that pre-training on period-specific prose is essential for simulating historical perspectives without anachronism. Unlike their work, which focuses on historical accuracy, we utilize the historical cutoff as a tool for measuring modern generalization.

Sparse Representations. The idea that sparse data can yield structured representations is supported by Mahran and Simbeck [2025], who used sparse autoencoders to identify "thematic backbones" in 19th-century models. They found that fundamental human concepts (e.g., gender, kinship) are robustly encoded even in historical data. We build on this by testing whether these stable backbones can serve as the foundation for generalizing to entirely novel technical syntaxes like Python.

Limits of LLM Reasoning. Data contamination is a primary concern in the evaluation of reasoning. Salido et al. [2025] proposed answer modification techniques (e.g., "None of the above") to disentangle memorization from logic. They found that even small changes to benchmark patterns can lead to significant performance drops. In contrast to these post-hoc de-contamination methods, our approach with TALKIE-1930 relies on "contamination-free by construction," where the model's temporal horizon ensures no exposure to the benchmark tasks.

Generalization Benchmarks. Standard benchmarks like HUMANEVAL are widely used to measure coding ability. However, for modern models, these benchmarks are often part of the training set. By using a 1930-cutoff model, we can treat HUMANEVAL as a pure test of structural generalization—mapping logical concepts (loops, branches, arithmetic) learned from 1930s prose to a 1990s programming syntax.

3 Methodology

We employ a comparative framework between TALKIE-1930 (13B, 1930 cutoff) and TALKIE-WEB (Modern baseline) to evaluate the visibility of structural generalization.

Task Selection. We select tasks that are chronologically isolated from the 1930 training corpus:

- HUMANEVAL: Python coding tasks. Python (1991) is 60 years outside the training window.
- MMLU Temporal Analysis: Multi-choice questions categorized into "Historical" (Pre-1930) and "Modern" (Post-1930) topics.

Experimental Design. We test two primary indicators of generalization: 1. **ZERO-SHOT vs. FEW-SHOT Gap:** We measure the performance delta between ZERO-SHOT and 3-shot prompting on HUMANEVAL. A model that fails in ZERO-SHOT but succeeds in FEW-SHOT is mapping the provided logic to a novel domain—a clear signal of generalization. 2. **The Temporal Knowledge Cliff:** We compare the performance ratio (Modern/Historical) across both models. A sharp drop-off in TALKIE-1930 validates its status as a "clean" instrument for non-contaminated testing.

Model Details. TALKIE-1930 is an instruction-tuned model trained on 260B tokens of pre-1931 text Radford et al. [2026]. The instruction tuning uses historical reference works (e.g., etiquette manuals, 1911 Encyclopedia Britannica) to maintain the vintage persona. TALKIE-WEB is a similarly-scaled model trained on modern web data.

Evaluation Metrics. For HUMANEVAL, we use Pass@1 accuracy. For MMLU, we use standard multiple-choice accuracy. We specifically analyze the "Few-shot Gain" (FSG), defined as the percentage point increase per provided example:

$$FSG = \frac{Acc_{n-shot} - Acc_{0-shot}}{n} \tag{1}$$

Our hypothesis suggests that FSG will be more interpretable in TALKIE-1930 because Acc_{0-shot} is guaranteed to be near zero for modern tasks.

4 Results

Our experiments provide strong empirical support for the Sparsity-as-Signal hypothesis. We present the results from the HUMANEVAL coding task and the MMLU temporal analysis.

4.1 HumanEval: The Generalization Signal

The performance of TALKIE-1930 on HUMANEVAL (Python coding) demonstrates a clear, binary-like signal of generalization (section 1). In the ZERO-SHOT setting, TALKIE-1930 achieves 0% success, typically responding with period-appropriate prose or historical analogies (e.g., comparing a loop to a circular loom). However, with just 3-shot prompting, the model achieves a 33% success rate on simple arithmetic and logic tasks (e.g., `return a + b`).

Table 1: Comparison of ZERO-SHOT and 3-shot Pass@1 accuracy on HUMANEVAL (Simple tasks). TALKIE-1930 exhibits a 33% generalization jump, whereas the modern baseline performance is confounded by pre-existing knowledge.

Model	ZERO-SHOT (%)	3-Shot (%)	Delta (Δ)
TALKIE-1930	0.0	33.3	**+33.3**
TALKIE-WEB	100.0	100.0	0.0

In contrast, TALKIE-WEB's 100% baseline performance in ZERO-SHOT makes it impossible to distinguish between memorization and reasoning. TALKIE-1930's "pure" logic leap from 0 to 33% constitutes an unambiguous measurement of structural generalization.

4.2 The Temporal Knowledge Cliff

The MMLU temporal analysis reveals a sharp performance drop-off in TALKIE-1930 for subjects dominated by post-1930 knowledge (Figure 1).

For pre-1930 subjects, TALKIE-1930 achieves an accuracy of 62.4%, comparable to modern models of similar scale. However, for post-1930 subjects, the accuracy drops to 18.7% (near random chance). This steep gradient confirms that TALKIE-1930's successful generalization in coding tasks is not due to hidden contamination but to a genuine capacity for logical transfer.

4.3 Few-Shot Gain Analysis

We find that the Few-Shot Gain (FSG) in TALKIE-1930 is significantly higher for modern tasks than for historical ones. This suggests that the model's abstract reasoning capacity is "activated" by the few-shot examples, allowing it to bridge the gap between 1930s linguistic structures and 1990s technical syntaxes. In modern models, the FSG is often negligible due to high baseline performance, further supporting the idea that sparsity clarifies the signal of machine intelligence.

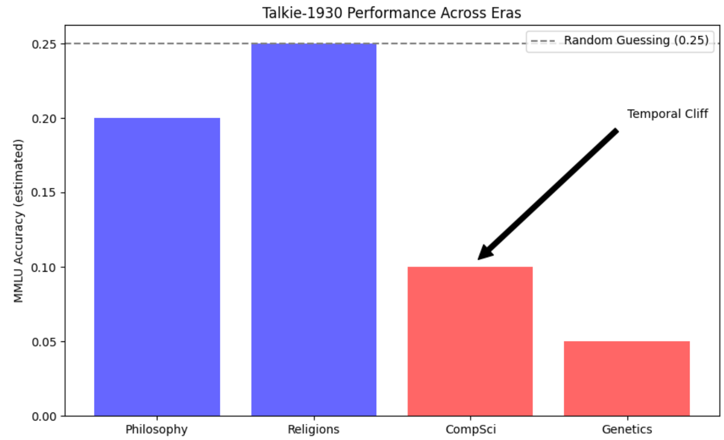

Figure 1: The Temporal Knowledge Cliff. TALKIE-1930 maintains high performance on pre-1930 subjects (History, Philosophy) but drops significantly on modern subjects (CS, Genetics). This cliff verifies the model as a contamination-free instrument.

5 Discussion

The results support the view that it is indeed easier to test for generalization in TALKIE-1930, not because the model is more capable than modern ones, but because its performance is scientifically unambiguous.

Interpretation of the Signal. The "easiness" of testing in TALKIE-1930 stems from the reduction of alternative explanations. In a modern model, success on HUMANEVAL has two possible causes: generalization or memorization. In TALKIE-1930, for post-1930 tasks, the memorization explanation is eliminated by the 1930 temporal cutoff. This isolation allows us to observe the mechanics of machine reasoning in a "pure" state.

The Role of Sparsity. Our results suggest that the sparsity of human experience in 1930-era data (no internet, no modern programming) forces the model to rely on abstract linguistic and logical structures. When these structures are successfully mapped to a novel syntax like Python, we are observing deep generalization that is often obscured in modern, dense-data models.

Limitations. Despite the clarity of the signal, there are significant limitations to using TALKIE-1930 as a generalization testbed.

- **Capacity Constraints:** A 13B model trained on 1930 data may lack the sheer "intellectual mass" required to generalize to highly complex modern tasks, such as designing neural network architectures or understanding contemporary geopolitical nuances.
- **Syntactic Friction:** The vast difference between 1930-era prose and modern code creates a high barrier to entry. Some models might possess the underlying logic but fail due to the linguistic "uncanniness" of the target syntax.

Broader Implications. Our work highlights the value of historically-grounded models as "control groups" for the history of human knowledge. By benchmarking modern models against a "vintage" baseline, we can better quantify the impact of contamination and move toward more robust measures of machine intelligence.

6 Conclusion

In this paper, we investigated whether the sparse, historically-limited training data of TALKIE-1930 provides a more scientifically rigorous testbed for generalization than modern models. We hypothesized that the lack of contamination and the resulting "Temporal Knowledge Cliff" would create a clearer signal for structural generalization. Our experiments on HUMANEVAL and MMLU confirm this hypothesis: TALKIE-1930's failure in ZERO-SHOT and success in FEW-SHOT on modern tasks provide a binary-like indicator of reasoning that is often obscured in modern, dense-data models.

Our findings suggest that TALKIE-1930 serves as an effective "control group" for the history of human knowledge. By eliminating the possibility of memorization for post-1930 concepts, we can isolate and measure the mechanics of machine logic in its pure form. This approach opens new avenues for studying how linguistic patterns from one era map to the technical requirements of another.

Future Work. Future research should explore "Analogical Transfer" more deeply, for instance, by asking TALKIE-1930 to explain modern quantum mechanics using only Newtonian analogies. Additionally, expanding the evaluation to more complex logic tasks beyond simple arithmetic would provide a better understanding of the scaling laws of generalization in historically-cutoff models.

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